

Predictive Model Report

Supermarket Sales Dataset, Customer Type Prediction

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Task: Task 2, Predictive Model Building

Dataset: Supermarket Sales Dataset (Kaggle)

Tools used: Python, Pandas, Scikit-learn, Matplotlib

1. Introduction

For this task I used the same Supermarket Sales dataset from my Task 1 analysis. This time, instead of just describing what's in the data, I wanted to build something that actually predicts something: whether a customer is a Member or a Normal customer, just from the details of one transaction. I picked this because it felt like a real question a store might actually care about. If you could tell from someone's shopping that they might sign up for membership, you could offer it to them right then.

2. Approach

1. I picked nine columns to predict from: Branch, City, Gender, Product line, Unit price, Quantity, Total, Payment method, and Rating.
2. Models can't read text, so I turned the text columns (Branch, City, Gender, Product line, Payment) into numbers using label encoding.
3. I split the data into 800 rows to train on and 200 rows to test on, so I could check the model on data it hadn't already seen.
4. I went with a Random Forest Classifier. It's basically a bunch of decision trees voting together on the answer, and it usually works decently well without needing much tuning.

3. Results

The model got 52% accuracy and an F1 score of 0.515 on the test set. Since there are only two categories and they're roughly split evenly, just guessing randomly would already get you close to 50%. So honestly, the model barely did better than flipping a coin.

	Precision	Recall	F1-score	Support
Member	0.50	0.55	0.52	97
Normal	0.54	0.50	0.52	103

It did about the same on both groups. It caught Members slightly more often (recall of 0.55), and was a bit more accurate when it guessed Normal (precision of 0.54). Either way, it wasn't confident about either one.

4. Which Features Mattered Most

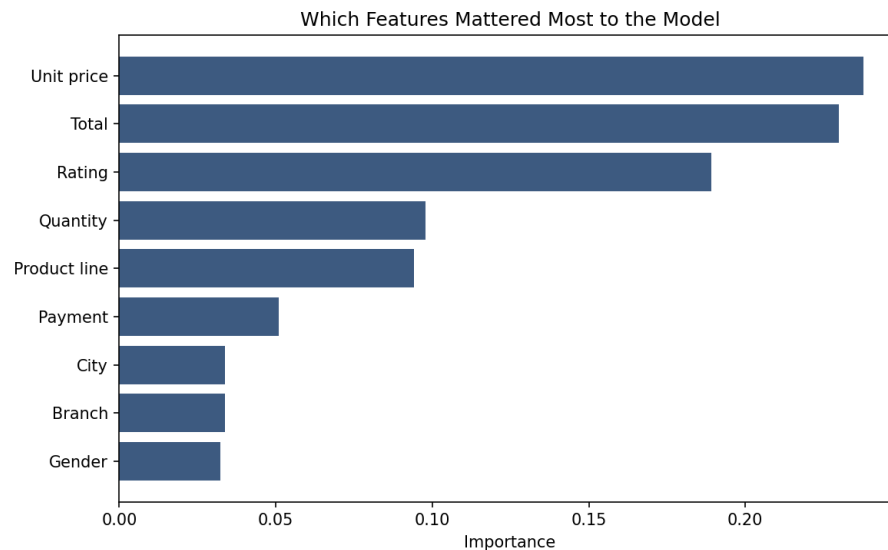


Figure 1: Feature importance ranking from the trained model.

Unit price (0.238), Total (0.230), and Rating (0.189) were what the model leaned on most, together making up over half of what it based decisions on. What surprised me a bit is that Gender, Branch, and City ranked at the bottom, each only around 3 to 4 percent. So if there's any real signal in here at all, it seems to come from spending and ratings, not from who someone is or where they shopped.

5. Limitations and Biases

The model only reached 52% accuracy which is barely better than a random guess, which tells me shopping behavior alone isn't a strong predictor of whether a customer is a Member or Normal customer in this dataset. A few reasons likely explain this.

To begin with, the dataset is relatively small at 1,000 rows, and only 800 of those were available for training. A larger dataset might reveal patterns that aren't visible here.

Also, the features I used (branch, product line, spending, payment method) describe a single transaction, not a customer's full shopping history. Membership status is probably shaped by factors this dataset doesn't capture, like how often someone shops, how long they've been a customer, or whether they were specifically offered membership at checkout.

Futhermore, Unit price, Total, and Rating were the features the model relied on most, but these still weren't strong enough to produce confident predictions, suggesting spending amount alone doesn't clearly separate 'Members' from 'Normal' customers.

A possible bias worth noting is that this dataset only covers three months from one supermarket chain, so any patterns found here may not generalize to other stores, regions or time periods.

6. Conclusion

Even though the model didn't perform well, I think this was still a useful thing to try. Not every question has a clear answer sitting in the data, and I think it's worth saying that plainly instead of pretending the model worked better than it did. If I picked this up again, I'd want data on each customer's full purchase history instead of just one transaction at a time, since that seems way more likely to actually show who's a Member and who isn't.